



FINAL YEAR CAPSTONE PROJECT

PHASE 1 – REVIEW 1

AY 2025–26

MandiSense AI

An Agentic Multi-Model Framework for Volatility-Aware
Agricultural Market Price Forecasting & Decision Intelligence

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Presentation Agenda

01 Introduction

- Problem Context & Background
- Problem Statement
- Motivation & Need

02 Literature Survey

- Overview of Existing Methods
- Hybrid Forecasting & Time Series
- Volatility & Risk Analysis
- Ensemble & Regime Detection
- Anomaly & Multi-Commodity
- Smart Agri., XAI & Surveys
- Identified Research Gaps

03 Proposed Work

- Project Objectives
- System Architecture & Solution
- Domain Knowledge Integration

04 Conclusion

- Beneficiaries & Impact
- Research Paper Details
- References

Problem Background

Context of the Problem

- India's agri. market (~INR 52 lakh crore) spans **7,000+ mandis**
- Prices driven by supply-demand, weather, festivals, MSP & export bans
- **58% of rural India** depends on agriculture — no forecasting exists

Real-World & Academic Relevance

- Onion **tripled in 45 days** (2023); tomatoes spiked **400%**
- Agmarknet, eNAM are **backward-looking** — zero prediction
- Existing ML: **monolithic, static, opaque** (no XAI)

Research Need & Significance

- 140M+ farmer households lose up to **30% income** to distress sales
- **No framework** unifies forecasting + GARCH + regime + XAI + cross-commodity
- Need: **predictive, volatility-aware, interpretable** mandi-level system

Who Is Affected

- **Farmers:** Sell at rock-bottom due to zero foresight
- **Traders:** No volatility alerts for procurement
- **Policymakers:** Reactive crisis management

Problem Statement

! The Problem

Indian mandi prices are **volatile and unpredictable** due to weather, seasonal gluts, and policy shifts. Current tools are **monolithic, static, and opaque** — providing no reliable or explainable price intelligence.

⚙️ Scope & Constraints

- Multi-agent mandi-level forecasting
- Volatility + regime detection
- Explainable via SHAP/LIME
- Cross-commodity spillover
- **7-day ahead**, daily granularity

🔑 Expected Outcomes

- **MAPE <15%** across regimes
- Real-time 2σ deviation alerts
- SHAP driver attribution
- Reduced farmer uncertainty
- Scalable across commodities

Motivation & Need

22/25

monolithic models

0/25

adaptive ensembles

0/25

unified framework

18%

agri. GDP share

📍 Why Forecasting Matters & Who Benefits

- **7,000+** mandis with unique dynamics
- 2023 onion/tomato crisis: mass losses
- No forecasting infrastructure exists
- **58%** of rural India in agriculture
- **Farmers:** Early price foresight
- **Traders:** Volatility alerts
- **Policy:** Data-driven intervention
- **Research:** Multi-agent benchmark

💡 Core Motivation

Information asymmetry causes **massive farmer losses** every cycle. A **unified, volatility-aware, explainable** framework empowers all stakeholders.

Literature Survey Overview

Econometric

- **ARIMA/SARIMA:**
Linear & seasonal trends
- **GARCH Family:**
Volatility clustering
- **HAR + LASSO:**
Realized volatility
- **TVP-VAR:** Policy uncertainty

🚫 **Limit:** Poor regime adaptation; fails in non-linear shifts

Deep Learning

- **LSTM/GRU:**
Temporal dependencies
- **Transformers:**
Long-range attention
- **CNN-Hybrid:**
Spatial-temporal feats
- **Decomposition:**
VMD/Ens. noise removal

🚫 **Limit:** Black-box opaque; implicit volatility only

Hybrid & Ensemble

- **ARIMA-LSTM:**
Residual modeling
- **Optimization:**
PSO/CS weight tuning
- **Stacking:**
Meta-learner combination
- **HMM-Guided:**
Regime-aware switching

🚫 **Limit:** Weights often static; not fully adaptive

Literature Survey – Hybrid Forecasting & Time Series

Theme	Author & Year	Method / Technique	Dataset / System Context	Key Contribution	Critical Insight / Takeaway
Hybrid Crisis Forecasting	Shah et al., 2025	ARIMA, LSTM, GBM, VaR	9 crisis periods; agri. commodity futures	Compared models across crisis episodes; LSTM best in volatile periods	Models used independently; no ensemble; no real-time anomaly detection
ARIMA-LSTM Hybrid	Kumar et al., 2023	ARIMA, LSTM, RF, GARCH	Monthly pulse prices (gram, moong, urad)	Residual hybrid improved RMSE by 8–25%; RF for lag selection	Monthly data only; no multi-agent design; volatility handled indirectly
Dual-Decomposition DL	Sinha et al., 2025	ICEEMDAN, VMD, GRU	UK fruit price datasets	Noise removal before GRU; outperforms LSTM/RNN variants	High complexity; no volatility or anomaly module; single pipeline
Mandi-level Prediction	Meena & Chaitra, 2024	VAR, Stacked LSTM, GRU, 1DCNN	Ragi prices, Karnataka mandis	MAPE ~4%; VAR captures cross-mandi dependencies	No ensemble or volatility module; single commodity; static weights

Literature Survey – Volatility Modeling & Risk Analysis

Theme	Author & Year	Method / Technique	Dataset / System Context	Key Contribution	Critical Insight / Takeaway
Realized Volatility Forecasting	Fiszeder et al., 2024	LASSO, HAR-RV	Agricultural futures markets	LASSO-HAR outperforms news-based indicators for volatility	Volatility only — no price forecasting; no decision system integration
Financial Stress & Volatility	Algirdas & Demir, 2024	HAR-RV variants, realized moments	Agri. commodity volatility data	Financial stress shows in-sample predictability; financialization supported	Poor out-of-sample forecast; no adaptive or real-time model
DL Volatility Prediction	Le et al., 2024	Bi-LSTM, GARCH, Adam	Cotton price data (AgTech)	91.38% volatility accuracy; denoising improves performance	Volatility only — no price-level forecast; no regime detection or XAI
Policy Uncertainty Impact	Wang et al., 2024	TVP-VAR	China agri. commodity prices	Time-varying policy transmission on price volatility captured	Country-specific; no direct forecasting framework; limited applicability

Literature Survey – Ensemble, Optimization & Regime Detection

Theme	Author & Year	Method / Technique	Dataset / System Context	Key Contribution	Critical Insight / Takeaway
Swarm-Optimized Ensemble	Chen et al., 2023	PSO, Cuckoo Search, EWT, ARIMA, BPNN	Corn and wheat futures	Swarm-based weight optimizer reduces MAPE by 43.66%	Centralized ensemble; no XAI or volatility alerts; high computational cost
HMM-Guided DL	Goswami et al., 2024	HMM, LSTM, CNN, GRU, BiLSTM	Potato prices, West Bengal	HMM detects market regimes; feeds signal into DL models	Region-specific; no ensemble or multi-agent framework
Anomaly Detection (MTS)	Zhao et al., 2024	Transformer (VTT), self-attention	Multivariate time series	Temporal + variable attention captures dependencies; anomaly interpretation	Not applied to agriculture; no price forecasting integration

Literature Survey – Anomaly Detection & Multi-Commodity

Theme	Author & Year	Method / Technique	Dataset / System Context	Key Contribution	Critical Insight / Takeaway
Multi-Commodity Forecasting + Anomaly	Annamalai et al., 2025	Transformer, LSTM-VAE, attention	Multi-commodity agri. prices	Forecasting + anomaly detection; captures sudden price spikes	Centralized; no agent-based design; no volatility or risk quantification
Smart Farming ML	Mistry et al., 2025	XGBoost, RF, KNN, SVM, Linear Reg.	Crop recommendation datasets	Crop recommendation + price prediction using ML ensemble	No time-series dynamics; static recommendation; limited interpretability

Literature Survey – Smart Agriculture, XAI & Surveys

Theme	Author & Year	Method / Technique	Dataset / System Context	Key Contribution	Critical Insight / Takeaway
Federated XAI	Tahir et al., 2025	FL, CNN, LSTM, SHAP, LIME, Grad-CAM	IoT-based smart agri. data	FL preserves privacy; SHAP+LIME+Grad-CAM for interpretability	No price forecasting; complex federated setup; no multi-agent design
Smart Agri. Survey	Kaur et al., 2024	IoT, CNN, XAI, ARIMA, edge computing	Precision farming systems	Comprehensive survey of IoT, edge, XAI for agriculture	Policy-focused; no forecasting; no multi-agent or volatility component
Deep Ensemble Survey	Survey, IEEE, 2024	PSO, SVM, RF, Bi-LSTM, CNN	Crop yield datasets	Survey of ensemble methods for crop recommendation	Survey only; no novel model; limited to yield, not price forecasting

Literature Synthesis: Key Findings

👍 Strengths & Trends Observed

- **T1:** Hybrids cut RMSE **8–57%** vs. standalone
- **T2:** GARCH volatility clustering (**91%** acc.)
- **T3:** Transformers beat RNNs on long range
- **T4:** XAI (SHAP/LIME) boosts trust
- **T5:** ICEEMDAN/VMD effective denoising

🗨️ Critical Limitations Identified

- **22/25** use **monolithic** pipelines
- **0/25** use **adaptive** ensemble weights
- Volatility = **output**, not predictive signal
- XAI & forecasting in **separate silos**
- **Zero** cross-commodity modeling

💡 **Key Insight:** The gap is **architectural, not algorithmic**. GARCH, HMM, SHAP, Transformers exist individually, but **no framework unifies** them into a single multi-agent system.

Research Gaps Identified

Gap 1: Monolithic Architectures

Single-model pipelines cannot model heterogeneous market drivers independently.

Gap 2: Static Ensembles

Fixed weights treat stable markets and crises identically—unreliable when accuracy matters most.

Gap 3: Volatility as Afterthought

Volatility treated as output metric, not integrated into forecasting logic or adaptive weighting.

Gap 4: No Regime Awareness

Models assume stationarity; fail to detect structural breaks from monsoon surplus or festival spikes.

Gap 5: No Explainability

Black-box outputs erode trust; no driver attribution or interactive interpretability.

Gap 6: No Cross-Commodity Analysis

All models forecast single commodities; ignore spillover and substitution effects.

★ **Core Gap:** No framework simultaneously integrates **multi-agent + adaptive ensemble + volatility + regime-switching + XAI + cross-commodity intelligence**.

Comparative Gap Analysis

Study	Multi-Ag.	Adapt.Ens.	Volatility	Regime	XAI	Cross-C.
Shah '25	×	×	Partial	×	×	×
Kumar '23	×	×	Partial	×	×	×
Sinha '25	×	×	×	×	×	×
Fiszeder '24	×	×	✓	×	×	×
Chen '23	×	×	×	×	×	×
Goswami '24	×	×	✓	✓	×	×
Le '24	×	×	✓	Partial	×	×
Annamalai '25	×	×	×	×	×	Partial
Tahir '25	×	×	×	×	✓	×
Meena '24	×	×	×	×	×	×
MandiSense AI	✓	✓	✓	✓	✓	✓

✔ **MandiSense AI is the only framework addressing all six dimensions simultaneously**

Project Objectives

Multi-Agent Engine

Gap 1/2: Monolithic & Static Models

- › **Agents:** Season, Arrival, Sentiment
- › **Method:** Adaptive Ensemble
- › **Target:** MAPE <15% (7-day)

Volatility System

Gap 3/4: Ignored Volatility & Regimes

- › **Feat:** Real-time GARCH Gauge
- › **Alerts:** 2σ Deviation Trigger
- › **Target:** Accuracy >85%

LLM Decision Intel

Gap 5: No Interpretability

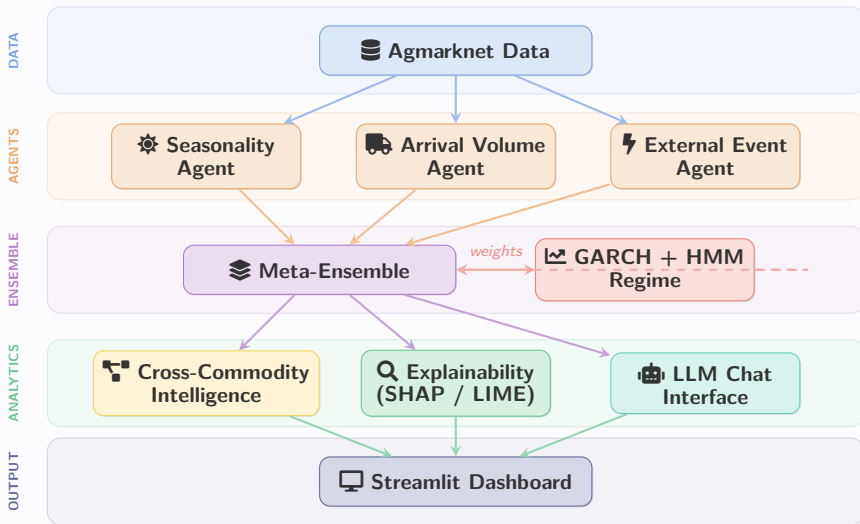
- › **Feat:** Ask-Your-Data (CSV)
- › **Tech:** Gemini API + CodeGen
- › **Target:** <10s, 95% success

Cross-Commodity Intel

Gap 6: No Cross-Commodity Analysis

- › **Feat:** Spillover & substitution
- › **Tech:** Granger Causality + VAR
- › **Target:** Lag <3 days, acc >85%

Proposed System Architecture



Technology Stack & Deployment

Core Stack

Python 3.10+
Pandas, NumPy
Jupyter Notebooks

ML / DL

Scikit-learn, XGBoost
TensorFlow / Keras
LSTM, GRU agents

Econometrics

arch (GARCH/EGARCH)
hmmlearn (HMM)
statsmodels (VAR, Granger)

XAI & LLM

SHAP, LIME
LangChain + Ollama
RAG on private data

Dashboard

Streamlit
Plotly, Matplotlib
Interactive charts

Data Source

Agmarknet (CSV/API)
Multi-commodity daily
5+ years historical

Domain Knowledge & Computing Solution



Theoretical Foundations



Agricultural Economics:

Mandi-level price formation, MSP policy, supply-demand, seasonal cycles



Time-Series Theory:

Non-stationarity, autocorrelation, decomposition, lag selection



Volatility Theory: GARCH

captures heteroskedasticity; EGARCH models asymmetric leverage



Regime-Switching: HMM detects structural breaks — latent states govern transitions



Computing Solution Design



Multi-Agent Systems:

Autonomous agents (seasonality, arrival, external) decompose the problem into specialized sub-tasks



Adaptive Ensemble: Meta-learner dynamically re-weights agents based on regime signals from HMM



Explainable AI: SHAP for global feature attribution; LIME for local instance-level interpretability



Cross-Commodity: Granger causality tests + VAR for spillover detection between related crops

⚙️ **Justification:** Multi-agent → Gap 1, adaptive ensemble → Gap 2, GARCH → Gap 3, HMM → Gap 4, SHAP/LIME → Gap 5, Granger/VAR → Gap 6.

Who Benefits



Farmers

- Short-term price foresight **before selling**
- Reduce distress sales and improve income stability
- Transparent explanations of price predictions



Policy Makers

- **Early volatility alerts** for proactive market stabilization
- Data-driven intervention strategies during crises
- Cross-mandi comparative intelligence



Traders & Supply Chains

- Better procurement planning and inventory control
- Data-driven market intelligence for pricing decisions
- Volatility alerts for risk-aware trading



Researchers

- Reproducible multi-agent ensemble framework
- Benchmark for agricultural market AI
- Extensible architecture for new commodities and regions

Research Paper – Overleaf Link

Paper Structure & Flow

-  **Abstract** – Multi-agent volatility-aware framework overview
-  **Introduction** – Problem definition, research need, scope, constraints
-  **Related Work** – 5 thematic dimensions, 16 papers analyzed in depth
-  **Gap Analysis** – 6 research gaps + comparative table (10 studies)
-  **Proposed Solution** – Multi-agent architecture, objectives, computing approach
-  **Conclusion** – Summary + future work
-  **References** – 20 citations (IEEE format, 2023–2025)

Paper Details

Format: IEEE Two-Column Journal

Length: Minimum 3 pages

Citations: 20 references (2023–2025)

Keywords: Agricultural price forecasting, GARCH, multi-agent systems, regime-switching, explainable AI, cross-commodity intelligence

Overleaf Link:

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Thank You!

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Questions & Feedback Welcome